Triage AI

Business Analysis 3.2 (AIBUY3A)
Theme: An AI Solution for Industries | Group: Triage AI
Clinical workflow and resource optimization

Vaal University of Technology • Diploma in IT



Executive summary and theme alignment

Triage AI applies Fourth Industrial Revolution technologies to public hospital operations. The goal is proactive capacity planning and support for clinical staff.

Industrial challenge

Manual charting, casualty congestion and preventable 30-day readmissions constrain hospital capacity.

The 4IR solution

Machine learning, time series, deep learning, clinical NLP and an operational softbot.

Municipal value

Lower avoidable costs, better bed utilization and more capacity for local emergencies.

The public healthcare capacity crisis

Problem definition

Overcrowded casualty units, prolonged emergency waits, ICU bed shortages and avoidable 30-day readmissions put pressure on public hospitals. Administrative charting further limits time for bedside care.

Current state
Reactive bed allocation and understaffing during admission surges.

- Casualty bottlenecks
- Manual clinical documentation
- Limited discharge risk screening

Proposed state with Triage AI
Forecast demand, prioritize discharge follow-up and automate documentation support.

- Predictive staffing and bed planning
- Voice-assisted documentation
- Earlier identification of high-risk patients

Project assumption: charting consumes up to 35% of shift time; establish and cite the local baseline before deployment.

Business objectives and success criteria

15% fewer readmissions

First-year target: identify high-risk patients before discharge and arrange targeted follow-up.

20% shorter emergency waits

Target: use 14-week demand forecasts to plan staffing and bed availability.

25% less documentation time

Target: speech-to-text and medical entity extraction. Classifier success criterion: ROC-AUC > 0.65.

Earlier ICU deterioration alerts

Evaluate continuous vital signs. Forecast success criterion: MAE < 15 patients per week.

System architecture and data pipeline

End-to-end clinical support

EHR records, physiological telemetry and clinician dictation feed a common Python pipeline.

The operational softbot exposes capacity and triage information to hospital staff.

1 Data sources	UCI benchmark EHR records, ICU vital signs and doctor dictation.
2 Processing	Pandas cleaning, standard scaling, one-hot encoding and text tokenization.
3 Core AI engine	Random Forest, SARIMAX, PyTorch MLP, spaCy and pyttsx3.
4 User outputs	Discharge risk scores, a 14-week surge calendar, voice alerts and softbot responses.

Machine learning for 30-day readmission risk

Random Forest binary classifier

UCI 130-US Hospitals: 101,766 raw encounters. Clean duplicates, exclude hospice/expired records and use balanced class weights.

Reported recall 48.0% | ROC-AUC 0.64 | Target > 0.65 not yet met

True negatives

No early readmission; predicted low risk.

False positives

No early readmission; predicted high risk.

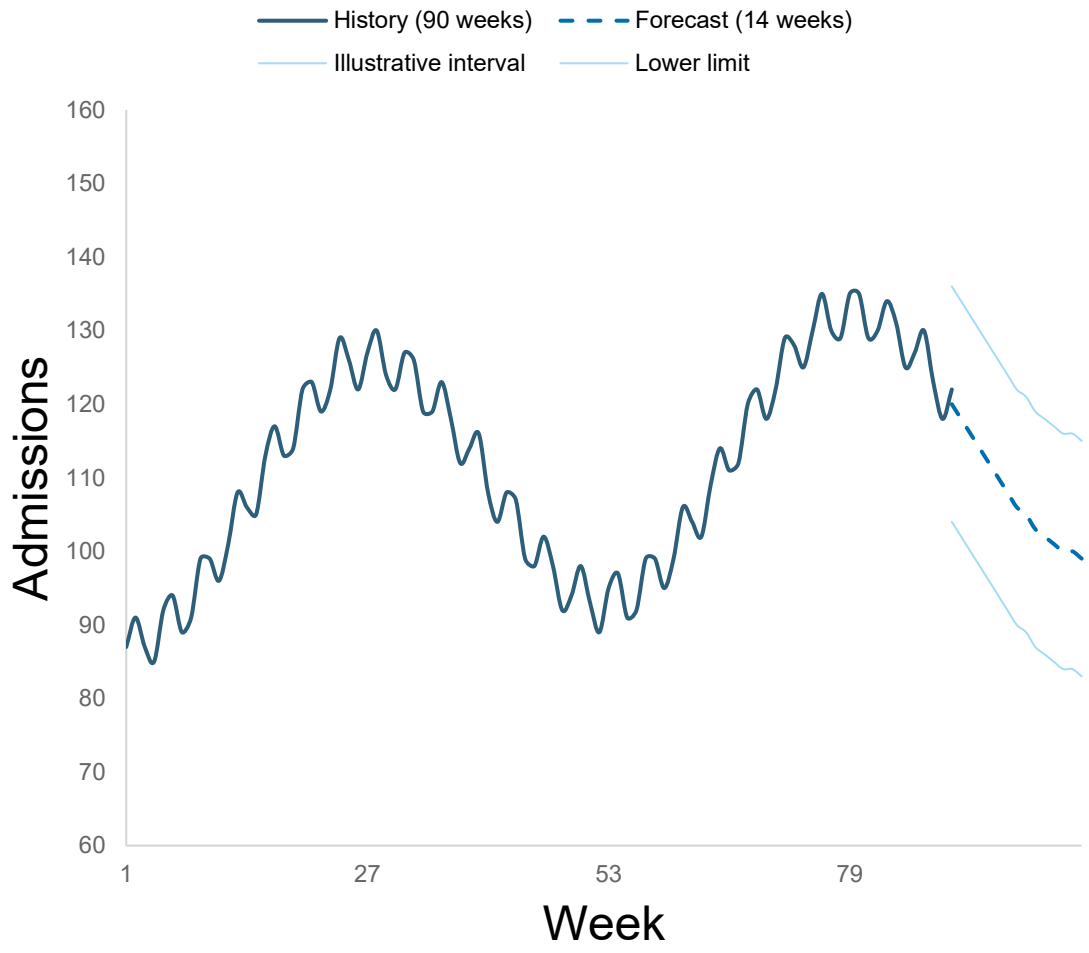
False negatives

Early readmission missed by the model.

True positives

Early readmission correctly flagged.

Emergency admission forecasting



SARIMAX models demand trends and annual seasonality. Chart values and uncertainty limits are illustrative; replace with the model output.

6.8

Reported MAE

patients per week

14

Forecast horizon

weeks

Deep learning for continuous ICU telemetry

PyTorch MLP

4 → 16 → 8 → 1

ReLU hidden layers and a Sigmoid output evaluate four vital-sign inputs.

Reported accuracy: 94.2% on simulated telemetry streams.

Example alert: risk score 0.88. This simulated output is not a validated sepsis probability.

- Heart rate 102 bpm
- MAP 78 mmHg
- SpO_2 91%
- Respiratory rate 23/min
- DETERIORATION ALERT
  Simulated demonstration

Clinical NLP and the operational softbot

Documentation and staff queries

Speech-to-text supplies a transcript for entity parsing. spaCy and regular expressions identify clinical concepts; pyttsx3 produces spoken operational alerts.

Clinical dictation example
“Acute dyspnea, chest tightness and peripheral edema. Furosemide 40 mg.”

- Symptom: dyspnea
- Symptom: chest tightness
- Medication: furosemide 40 mg

Operations softbot example
User: “What is current ICU bed capacity?”
Bot: “Ward B is 95% full; one bed remains.”

- User: “Protocol for Code Blue?”
- Bot: “Alert the resuscitation team.”
- Use the hospital-approved emergency contact.

Illustrative transcript and chat responses. Bed figures are simulated; emergency procedures must be configured for the hospital.

Governance constraints and risk mitigation

Development and deployment controls

Benchmark and simulated data support prototype development. A hospital deployment requires local governance, security review and clinical validation.

Data privacy and POPIA

Use de-identified benchmarks. Restrict access and review data handling before local deployment.

Class imbalance

Readmissions are a small minority. Use balanced class weights and assess recall with ROC-AUC.

Staff adoption

Reduce interaction burden through dictation and natural-language operational queries.

Model concept drift

Plan quarterly retraining and monitor performance against a held-out baseline.

Local healthcare and community impact

Financial efficiency

Prevent avoidable re-hospitalizations and reduce unnecessary bed occupancy. Expected benefit; quantify in a local pilot.

Emergency readiness

Target 20% shorter waits through predictive staffing and improved bed planning.

Clinician well-being

Target 25% less charting time and return more shift time to direct bedside care.

Questions & discussion

Triage AI | Business Analysis 3.2

github.com/BB-Dube/Triage-AI

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